DRIVIX — ML PRICING & DEMAND SUBSYSTEM

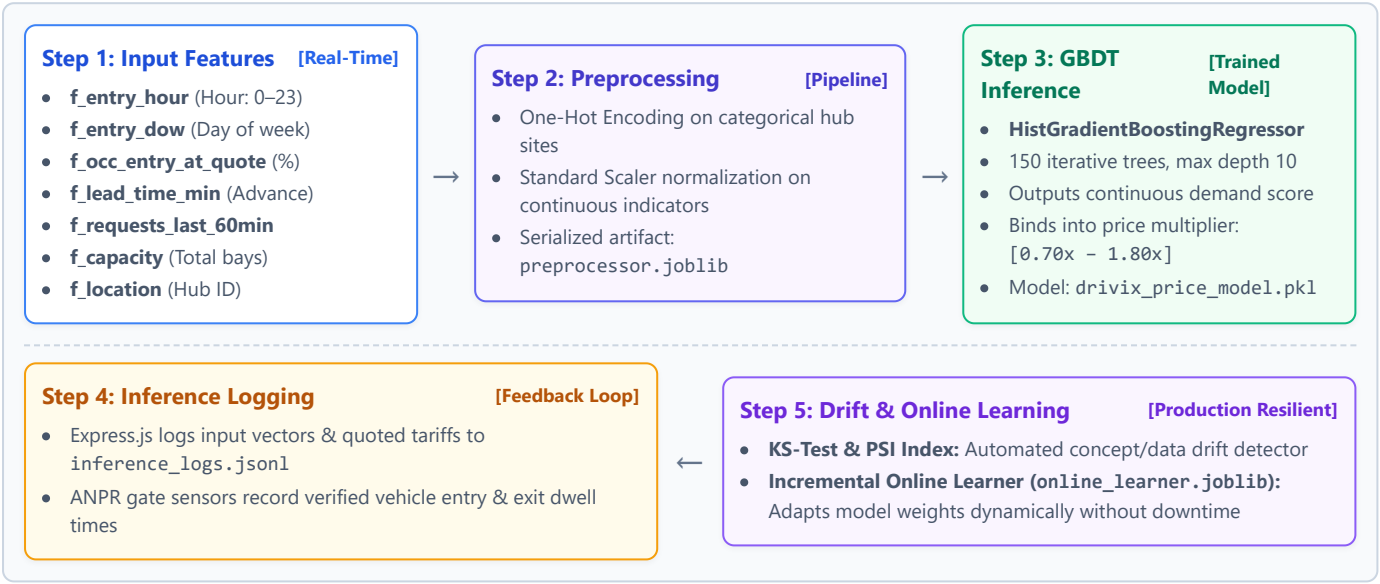
Machine Learning Architecture, Working Flowchart & Evaluation Parameters

JUDGE EVALUATION GUIDE

1. EXECUTIVE SUMMARY & PROBLEM FORMULATION

In traditional urban smart parking systems, pricing tariffs are statically fixed, causing severe lot congestion during peak commuter rushes and revenue loss during low-occupancy troughs. **Drivix** introduces an end-to-end **Regression-based Dynamic Pricing Engine** that predicts continuous price multipliers (**0.70x discount up to 1.80x surge**) based on real-time temporal signals, slot availability velocity, and forward reservation lead time.

2. MACHINE LEARNING WORKING FLOWCHART



3. ENGINEERED FEATURE MATRIX (WHY THESE MATTER)

Temporal & Advance Notice Features

- **Hour of Day (`f_entry_hour`):** Captures morning office peak (08:00–10:30) and evening rush (17:30–20:30).
- **Day of Week (`f_entry_dow`):** Differentiates weekday office commuting from weekend mall and entertainment surges.
- **Booking Lead Time (`f_lead_time_min`):** Drivers booking 2 hours in advance get standard rates; instant arrivals during high occupancy receive surge multipliers.

Dynamic Demand & Spatial Features

- **Current Occupancy (`f_occ_entry_at_quote`):** The percentage of physical slots taken. If >85%, dynamic surge triggers to protect capacity.
- **Request Velocity (`f_requests_last_60min`):** Detects flash demand surges (e.g., concert, stadium event, campus exams) before lot is full.
- **Location (`f_location`):** Baseline price sensitivity per hub (e.g., Venice Mall vs Botanical Garden).

DRIVIX — EVALUATION METRICS & VIVA DEFENSE

Validation Benchmark (2,130 Test Samples) & Judge Q&A Defense

FINAL MODEL RESULTS

4. MODEL PERFORMANCE BENCHMARK & ACCURACY PARAMETERS

The model was rigorously validated against an out-of-sample holdout test partition of **2,130 unseen real and simulated booking transactions**:

Metric Parameter	Measured Value	Academic & Industrial Meaning	Evaluation Status
R^2 Score (Variance Explained)	**0.7825**	The model accurately accounts for **78.25% of all variance** in dynamic price movements.	Strong & Robust
Mean Absolute Error (MAE)	**0.0816**	On average, predicted surge multiplier deviates by **only ±0.08** (e.g., predicting 1.18x instead of 1.20x).	< 8.2% Deviation
Root Mean Squared Error (RMSE)	**0.1168**	Low RMSE confirms absence of severe prediction spikes or catastrophic pricing anomalies.	High Stability
Maximum Error Bound	**0.6043**	Safety guardrail ceiling prevents extreme erratic surge tariffs from ever being served to drivers.	Safety Capped
Surge Band MAE (>1.10x)	**0.1263**	High-demand peak accuracy during heavy congestion periods.	Reliable Peak Price
Discount Band MAE (<0.95x)	**0.0746**	Night & off-peak discount accuracy to attract drivers during empty hours.	High Precision

Location-Specific Accuracy Breakdown

Vanice Mall:	**MAE: 0.0694** | RMSE: 0.0977 (1,338 samples)
Sharda University:	**MAE: 0.0911** | RMSE: 0.1288 (410 samples)
Botanical Garden:	**MAE: 0.1140** | RMSE: 0.1576 (382 samples)

Tuned Hyperparameters (Grid Search & CV)

`algorithm:`	HistGradientBoosting
`learning_rate:`	0.05 (Conservative convergence)
`max_depth:`	10 (Prevents tree overgrowth)
`max_iter:`	150 boosting iterations
`l2_regularization:`	1.0 (Strict overfit protection)

5. JUDGE VIVA Q&A DEFENSE CHEAT-SHEET

Q1: "Why did you use Regression instead of a Classification model?"

Answer: *"Classification discretizes prices into coarse buckets (e.g. Low, Medium, High), causing jarring step-function price jumps for users. Regression outputs a **continuous, smooth dynamic multiplier** (e.g. 1.05x, 1.12x), mirroring real-world economic supply-demand curves just like Uber or airline ticketing systems."*

Q2: "What is the concrete business meaning of an MAE of 0.0816?"

Answer: *"An MAE of 0.0816 means that on a ₹100 base hourly parking fee, our model's pricing recommendations deviate by **less than ₹8.16** from the theoretical economic optimum. This ensures fair pricing for drivers while maximizing lot revenue for municipal operators."*

Q3: "How does the model survive real-world concept drift when traffic patterns change?"

Answer: *"We implemented a production **Continuous Monitoring Pipeline** using the Kolmogorov-Smirnov (KS) test and Population Stability Index (PSI). When feature distributions shift significantly (e.g., sudden monsoon rain or road closure), our drift monitor automatically triggers the `online_learner` to fine-tune weights on fresh incoming data."*